

Magnetic Field of a Circular Current Loop

$$P = (0, \theta, z) = \mathbf{r}$$

$$Q = (a, \theta, 0) = \mathbf{r}'$$

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = (-a, 0, z) = -a\hat{\rho} + z\hat{\mathbf{z}}$$

$$R = |\mathbf{r} - \mathbf{r}'| = \sqrt{a^2 + z^2}$$

Biot-Savart law

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\boldsymbol{\ell} \times \mathbf{R}}{R^3}$$

For $d\boldsymbol{\ell}$, the length vector of the wire element in the direction of current. Since a current carried by a circular loop, $d\boldsymbol{\ell}$ can be expressed by

$$d\boldsymbol{\ell} = a d\theta \hat{\boldsymbol{\theta}}$$

Substitution:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{(a d\theta \hat{\boldsymbol{\theta}}) \times (-a\hat{\rho} + z\hat{\mathbf{z}})}{(a^2 + z^2)^{3/2}}$$

Cross product in cylindrical coordinates

$$\begin{aligned} (a d\theta \hat{\boldsymbol{\theta}}) \times (-a\hat{\rho} + z\hat{\mathbf{z}}) &= \begin{vmatrix} \hat{\rho} & \hat{\boldsymbol{\theta}} & \hat{\mathbf{z}} \\ 0 & a d\theta & 0 \\ -a & 0 & z \end{vmatrix} \\ &= (az d\theta)\hat{\rho} + (a^2 d\theta)\hat{\mathbf{z}} \end{aligned}$$

Therefore, for $0 \leq \theta \leq 2\pi$,

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \left(\int_0^{2\pi} \frac{az}{(a^2 + z^2)^{3/2}} d\theta \hat{\rho} + \int_0^{2\pi} \frac{a^2}{(a^2 + z^2)^{3/2}} d\theta \hat{\mathbf{z}} \right)$$

Notice that

$$\hat{\rho} = \cos \theta \hat{\mathbf{i}} + \sin \theta \hat{\mathbf{j}}$$

If we calculate the $\hat{\rho}$ part,

$$\begin{aligned} &\int_0^{2\pi} \frac{az}{(a^2 + z^2)^{3/2}} \hat{\rho} d\theta \\ &= \frac{az}{(a^2 + z^2)^{3/2}} \left[\int_0^{2\pi} \cos \theta d\theta \hat{\mathbf{i}} + \int_0^{2\pi} \sin \theta d\theta \hat{\mathbf{j}} \right] = 0 \end{aligned}$$

This is because $\hat{\boldsymbol{\rho}}$ is symmetric while θ changes from 0 to 2π .
Thus,

$$\mathbf{B} = \left(\frac{\mu_0 I}{4\pi} \frac{a^2}{(a^2 + z^2)^{3/2}} \int_0^{2\pi} d\theta \right) \hat{\mathbf{z}}$$

and therefore

$$\boxed{\mathbf{B} = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}} \hat{\mathbf{z}}}$$